

Aim: To write an Embedded C program for interfacing PIR Sensor using Arduino Uno and Proteus

objective:

1. To interface a PIR motion sensor with an Arduino Uno in the Proteus simulation software.
2. To write code that detects changes in infrared radiation levels caused by human movement.
3. To display the motion detection status in real time using a Proteus Virtual Terminal..

program:

```
int Status = 13; // Digital pin D6
int sensor = 12; // Digital pin D7

void setup() {
  Serial.begin(9600);
  pinMode(sensor, INPUT); // declare sensor as input
  pinMode(Status, OUTPUT); // declare LED as output
}

void loop() {
  long state = digitalRead(sensor);
  Serial.println(state);
  if(state == HIGH) {
    digitalWrite (Status, HIGH);
    Serial.println("Motion detected!");
    delay(1000);
  }
  else {
    digitalWrite (Status, LOW);
    Serial.println("Motion absent!");
    delay(1000);
  }
}
```

Output:

